

Jira Activity Analyzer

v0.2.40 Current-State Archive & Stable Deduplication V2

Implementation status: Automated validation and Windows packaging completed. Real company Jira and manual GUI acceptance were not executed.

Storage identity: current_state / model version 1 / Stable Hash policy version 2

Branch: feat/v0.2.40-current-state-archive-stable-dedup-v2

Executive Summary

v0.2.40 replaces the writable version-history archive path with a separate Current-State SQLite database. Every Jira Issue has at most one Snapshot, one compressed Full Fetch Payload, and one Sync State. Stable-equal reruns update check metadata and event deduplication without rewriting large payload data.

Coverage downgrade, incomparable scope, Partial results, Failed results, payload validation errors, and database transaction failures cannot overwrite a previously valid Issue state.

Safety Position

v0.2.39 databases are detected read-only and remain unmodified. No in-place migration, cleanup, rename, compaction, or automatic activation is performed. Jira access remains read-only; formal writes target only the explicitly configured local SQLite database.

Schema and Row Invariants

Table	Maximum / identity rule
database_metadata	One authoritative database identity record
source_objects	One per Jira Issue ID; Issue Key also unique
current_issue_snapshots	One per Source Object (primary key)
current_full_fetch_payloads	One per Source Object (primary key)
issue_sync_states	One per Source Object (primary key)
activity_events	Many; unique Source Object + event identity hash
database_run_state	Latest compact run summary only

The new schema does not create or use source_object_versions, history source_payloads, or source_import_refs.

Stable Projection V2

Authoritative business fields, comments, changelog, attachment metadata, relationships, parent/subtasks, optional Remote Links, custom fields, and Coverage markers participate.

Fetch, run, queue, evidence and local path metadata; names; schema; renderedFields; and Jira updated alone do not participate. Set-like arrays use stable identity ordering while ADF description arrays preserve semantic order.

Known calculated fields are excluded only after exact Field ID resolution from the same names map. Missing mappings warn. Ambiguous mappings exclude nothing and report AMBIGUOUS_VOLATILE_FIELD_MAPPING.

Coverage and Transaction Design

Coverage states are Disabled, Skipped, Failed, Partial, CompleteEmpty, and CompleteNonEmpty. Comparison returns Equivalent, Upgrade, Downgrade, Incomparable, or Invalid. Only Equivalent and Upgrade candidates may proceed.

Payload bytes are canonicalized, SHA-256 hashed, gzip compressed, decompressed, re-hashed, and parsed before a formal transaction. Snapshot, Payload, Sync State, and Activity Events commit together per Issue. Fault injection verifies rollback after Snapshot mutation and during event insertion.

Five Locked Decisions

1. Stable-equal/equivalent content does not rewrite Snapshot, Payload BLOB, payload_updated_at, or content_revision.
2. Volatile custom fields resolve by exact Field ID; ambiguity is fail-safe.
3. Coverage downgrade is blocked; Disabled and CompleteEmpty remain distinct.
4. No per-Issue Import Ref or payload/version history is accumulated.
5. v0.2.39 databases are read-only and are not migrated.

Diagnostics

Stage 5 creates sanitized current-state-save-diagnostics JSON under the established exports/full-fetch-results folder. It contains decisions, hashes, coverage comparison, changed paths and counts, but no token, Authorization header, full Description, Comment body, Changelog body, or raw Payload.

Automated Validation

Command	Result	Duration
npm.cmd run typecheck	Passed	7.4 s
npm.cmd run test:v0.2.40	Passed - 57 scenarios	2.3 s
npm.cmd run test:unit	Passed	16.9 s
npm.cmd run test:integration	Passed	1.46 s
npm.cmd run test:regression	Passed	1.46 s
npm.cmd run build	Passed	23.48 s
npm.cmd run test:ui	Passed - 9 routes x 5 viewports	76.0 s
npm.cmd run dist	Passed	78.2 s

Automated synthetic four-Issue validation passed. Real Jira Label/volatile-field rounds, manual Windows GUI acceptance, and long-duration packaged smoke were not executed.

The final app-only UI smoke passed after startup selection and legacy expectation updates. All failed attempts and retry durations remain visible in the execution ledger.

Known Warning

Vite reports a non-blocking renderer chunk-size advisory above 500 kB. No new runtime failure was observed.

Windows Artifacts

Setup

release/Jira Activity Analyzer Setup 0.2.40.exe

106,471,213 bytes

684064BB0DB33F40D522AED5478B667249FA1D5E93A1A7C77D4AD48AB0A5FFFE

Portable

release/Jira Activity Analyzer Portable 0.2.40.exe

106,241,230 bytes

A7626132FACF09069DE200EB81B57728C0CB971305B8C2A5AB24506827E2ADB6

Known Limitations and Remaining Risks

- Current-State intentionally retains no historical Snapshot or Payload rows.
- Stable Hash exclusion policy may need future tuning as Jira plugins change.
- Volatile-only raw response changes do not replace the authoritative archive.
- Deleted or inaccessible Jira history cannot be recovered by refetching.
- Real company Jira and Windows manual acceptance remain external gates.

Git Delivery

Branch: feat/v0.2.40-current-state-archive-stable-dedup-v2

Baseline: 08ff3e313f0a6d7475050e91270e96c84489af67

Delivery commit and push are completed after final repository review. No v0.2.40 tag or GitHub Release is created before real-machine acceptance.

Execution Time

The complete command, retry, phase, failed-duration and reconciliation record is stored in reports/JiraActivityAnalyzer_v0.2.40_execution_time_ledger.json.